

Spoken Tutorial – Understanding Tokens in Large Language Models

Title: Understanding Tokens in Large Language Models

Module	Artificial Intelligence Essentials
Episode	6. Understanding Tokens
Learning Objective	At the end of this tutorial learner will be able to 1. Explain what tokens are and why they are fundamental to Large Language Models.
Approx. Duration	3-4 min
Outline	• Introduction to Tokens • What is a Tokenizer? • Impact on Context Windows • Impact on Cost • Impact on Chunking • Summary • Assignment
Meta Tags	tokens, tokenization, LLM, Large Language Models, context window, API cost, chunking, AI
Pre-requisite Tutorial	Introduction to Large Language Models

Script

Visual Cue	Narration
Title slide with text: 'Understanding Tokens in Large Language Models' and a subtle background related to AI or data processing.	Welcome to this Spoken Tutorial on Understanding Tokens in Large Language Models.
Learning Objectives slide with bullet points: 'Explain what tokens are' and 'Understand their importance for LLMs'.	In this tutorial, you will learn about tokens. You will explain what tokens are. You will also understand why they are fundamental to LLMs.
A person sitting at a computer, with an open web browser showing a search bar or a simple web application.	Here, I am using a browser on a computer or mobile. We will also use an online tokenizer tool.
Prerequisites slide with text: 'Internet connection' and 'Familiarity with LLMs'.	To follow this tutorial, you must be connected to the internet. You should also be familiar with what LLMs are.

Illustration of a sentence transforming into small, distinct puzzle pieces or building blocks, with 'The' as one block and 'Tokenization' splitting into three blocks: 'Token', 'iz', 'ation'.	Imagine an LLM as a chef. It doesn't read a full recipe sentence at once. Instead, it breaks it down into small, distinct ingredients. These small pieces are called tokens. Let's see how a sentence becomes tokens. Open an online tokenizer tool, like the one from Hugging Face. Type the word "The" into the input box. Observe that it might be one token. Now type "Tokenization". Notice how it breaks into "Token", "iz", "ation". This shows that LLMs process text in these smaller, sub-word units. They don't see whole words as we do.
A visual representation of a dictionary splitting a phrase 'Hello, how are you today?' into individual tokens, with arrows pointing from the raw text to the tokenized output.	Now, let's understand the tool that performs this. Think of a tokenizer as a special dictionary. This dictionary knows how to split raw text into the tokens an LLM understands. It's like a translator for the LLM. Let's use our online tokenizer again. Type a simple sentence: "Hello, how are you today?" Observe how the tokenizer breaks it down. Notice punctuation marks like commas might also be separate tokens. The tokenizer ensures the LLM gets a consistent format. This is crucial for the LLM to process language effectively.
A visual metaphor of a brain with a limited 'memory bar' filling up with small 'token' blocks, illustrating the concept of a context window with a numerical limit like '8,192 tokens'.	So what does this mean for LLMs? Imagine your brain has a limited short-term memory for conversations. This memory can only hold a certain number of words at a time. For an LLM, this "short-term memory" is its context window, measured in tokens. An LLM can only process a fixed number of tokens at once. This is a hard limit, for example, 8,192 tokens. Any text beyond this limit is simply ignored. This includes your prompt and any retrieved documents. So, efficient token usage helps your LLM remember more.
A visual representation of a stack of small coin-like tokens, with a dollar sign or currency symbol, and a graph showing cost reduction with fewer tokens.	Next, let's consider cost. Think of tokens like currency for LLM services. Every time you use an LLM, you pay per token processed. More tokens mean higher costs. API providers charge based on the number of tokens sent and received. An optimized system, like GraphRAG, uses fewer tokens. This can reduce operational costs significantly. So, understanding tokens helps you save money.

An illustration of a large document being split into smaller, equally sized 'chunks' or 'boxes', each labeled with a 'token count' to ensure they fit a specific limit.	Finally, let's look at chunking. Imagine you're packing items into boxes, but each box has a weight limit. If you overfill a box, it breaks. Similarly, when we "chunk" documents for an LLM, each chunk has a token limit. When preparing documents for RAG systems, we break them into "chunks". These chunks must fit within the LLM's context window. Therefore, we measure chunk size in tokens, not words. This ensures that the LLM can fully process each piece of information. It helps avoid losing valuable context.
Summary slide with bullet points summarizing key takeaways about tokens, tokenizers, context windows, cost, and chunking.	With this, we come to the end of this tutorial. Let us summarise. We learned that LLMs process text using tokens. A tokenizer splits text into these sub-word units. Understanding tokens is crucial for context windows, API costs, and effective chunking.
A person at a computer screen, showing an online tokenizer tool with different text inputs and their tokenized outputs, suggesting experimentation.	Now as an assignment, please do the following. Find an online tokenizer for a different LLM, like OpenAI's. Experiment by typing various words and sentences. Observe how different tokenizers might split text slightly differently. Think about how this could impact context windows and costs.
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